

Rare Category Analysis for Complex Data: A Review

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Though the sheer volume of data that is collected is immense, it is the rare categories that are often the most important in many high-impact domains, ranging from financial fraud detection in online transaction networks to emerging trend detection in social networks, from spam image detection on social media platforms to rare disease diagnosis in medical decision support systems. The unique challenges of rare category analysis include (1) the highly skewed class distribution; (2) the non-separable nature of the rare categories from the majority classes; (3) data and task heterogeneity; and (4) the time-evolving property of the input data sources. This survey reviews state-of-the-art techniques used in complex rare category analysis, where the majority classes have a smooth distribution while the minority classes exhibit the compactness property in the feature space or subspace. Rare category analysis aims to identify, characterize, represent, and interpret anomalies that not only show statistical significance but also exhibit interesting patterns (e.g., compactness, high-order structures, showing in a burst). We introduce our study, define the problem setting, and describe the unique challenges of complex rare category analysis. We then present a comprehensive review of recent advances that are designed for this problem setting, from rare category exploration without any label information to rare category exposition that characterizes rare examples with a compact representation, from the representation of rare patterns in a salient embedding space to the interpretation the prediction results and providing relevant clues for the end-users' interpretation. Finally we discuss potential challenges and shed light on the future directions for complex rare category analysis.¹

CCS Concepts: • **Information systems** → *Data mining*;

Additional Key Words and Phrases: Rare category analysis, imbalanced data distribution, anomaly detection

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1 INTRODUCTION

The so-called era of big data is defined in part by the sheer volume of multimodal data that are being collected from various platforms (e.g., graphs, images, text). Yet, it is the rare occurrences—those data points that are significantly different from other observations in the data—that often are of greatest importance in many high-impact domains: from financial fraud detection in online transaction networks to emerging trend detection in social networks, from spam image detection on social media platforms to rare disease diagnosis in medical decision support systems. Anomaly detection refers to the process by which these rare occurrences are identified. Many techniques have been proposed to characterize anomalies in domains such as financial fraud detection [102], insider threat prediction [66], novelty detection [92], gene disease discovery [17], emerging trend tracking [28], network intrusion detection [58], astronomy [133], spam image detection [93], and rare disease diagnosis [117].

1.1 Anomalies vs. Rare Category Examples

The branch of data mining concerned with identifying anomalies or outliers² has a lengthy history. In 1980, Douglas M. Hawkins proposed the definition of outliers [47] that is shown in Definition 1.1. Since this time, the problem of anomaly detection or outlier detection has been generalized and studied in various contexts, such as those of high-dimensional numerical data [173], sequential data [27], time-series data [45], graph data [7], and financial data [36, 113]. This wide-ranging exploration has yielded many domain-specific terms for outliers, including novelties events, surprising changes, frauds outbreak.

Definition 1.1. Hawkins’s Definition of Outliers [47]

An outlier is an observation that differs so much from other observations as to arouse suspicion that it was generated by a different mechanism.

Though researchers have had tremendous success applying anomaly detection methods in various domains, not all anomalies detected are necessarily useful or relevant to the actual events of interest. In fact, most anomalies are uninteresting data points that either are drawn from a known distribution of noise or correspond to combinations of features that are irrelevant to downstream applications [109]. For instance, in Figure 1, we present a set of sky images captured by ground-based telescopes as part of the **Sloan Digital Sky Survey (program)**.³ According to the Sloan Digital Sky Survey 99.9% of its captured sky images (e.g., the top row of Figure 1) can be well explained based on known phenomena of the universe (e.g., discovered stars, comets, nebulae); only 0.1% of the images (e.g., the bottom row of Figure 1) are anomalies. Moreover, among the anomalies, 99% of the images (e.g., bottom left image in Figure 1) are of no interest to astronomers and are caused by diffraction spikes of satellite trails or artifacts of the telescope. Only 1% of the instances (a minuscule 0.001% of the Sloan Digital Sky Survey database) are useful and may lead to new scientific discoveries (e.g., the bottom right image in Figure 1). Here, we refer to anomalies that are not only statistically significant but also interesting as rare category examples and the problem of studying rare category examples as **rare category analysis**. Figure 2 illustrates the relationship among anomalies, uninteresting anomalies, and useful anomalies (i.e., rare category examples), where rare category examples are a subset of anomalies that are not only statistically significant but also compact in the feature space.

We provide the following general definition of rare category analysis: Given a dataset $\mathcal{D}$ of n examples that come from C distinct classes, without loss of generality, we assume that $\sum_{i=1}^n \mathbf{x}_i = \vec{0}$

²In this article, we use “anomalies” and “outliers” interchangeably.

³<http://www.sdss.org/iotw/archive.html>

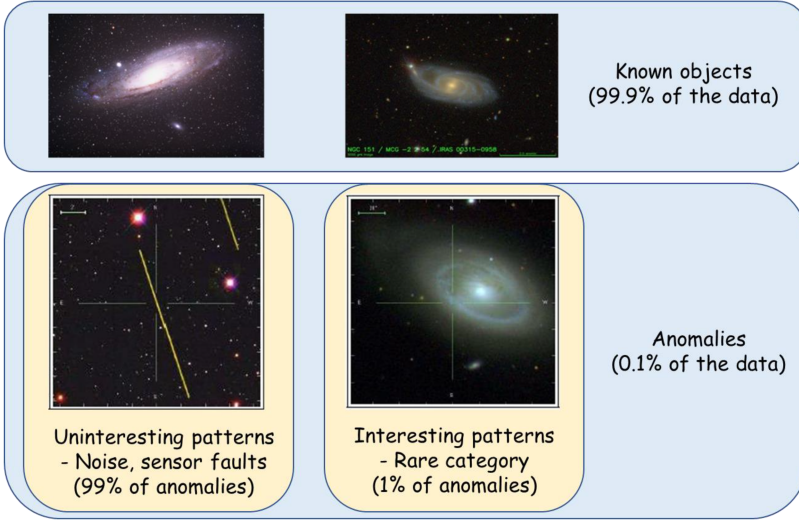


Fig. 1. Anomalies in Sloan data [109]. Top row: known objects. Bottom row: anomalies, 99% of which are uninteresting patterns, such as diffraction spikes (bottom left) and only 1% of which are interesting patterns that are worthy of future research and may lead to the discovery of extraordinary objects (bottom right).

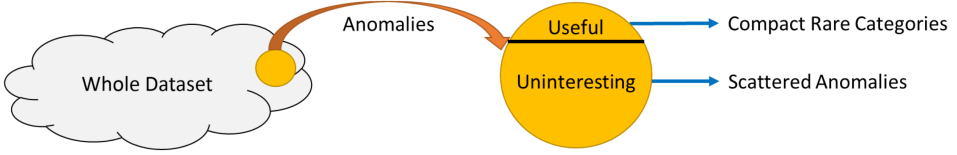


Fig. 2. Relationship between anomalies/outliers and rare category examples.

and $\frac{1}{n} \sum_{i=1}^n (x_i^j)^2 = 1$, where x_i^j denotes the j th feature of $\mathbf{x}_i$. For the sake of simplicity, we assume the class label for the i th node, $y_i = 0$, corresponds to the majority class (i.e., normal examples) and $y_i \in \{1, \dots, C-1\}$ corresponds to the rare categories (i.e., suspicious examples). Note that rare category analysis is the problem of studying patterns in the minority classes.⁴ In particular, for cases in which more than one majority class is observed, we treat all the majority classes as a single class once the underlying distribution of each majority class is sufficiently smooth [49]. In general rare category analysis, we make the following assumptions regarding the support region of the majority class and minority classes:

ASSUMPTION 1. Smoothness Assumption for Majority Class. *Given a highly imbalanced dataset, the distribution of the support region of each majority class is sufficiently smooth.*

ASSUMPTION 2. Compactness Assumption for Minority Class. *Given a highly skewed dataset, the minority class examples can be represented as a compact cluster in the feature space.*

These assumptions are made to ensure that the rare categories are identifiable and meaningful. To better capture this, let us look first at the example in Figure 3(a), where the majority class (shown in blue) has a Gaussian distribution with large variance on the left, while the minority

⁴In this article, we use “rare category” and “minority class” interchangeably.

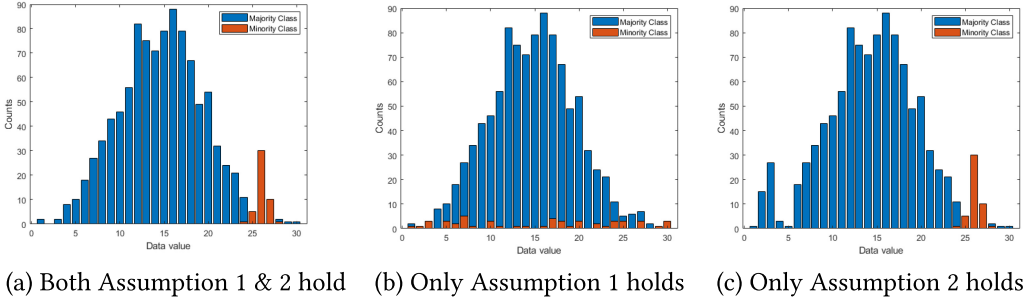


Fig. 3. The support regions of a majority class and a minority class in a one-dimensional synthetic dataset.

class (shown in orange) corresponds to a peak with small variance on the right. If the minority class is not compact and violates Assumption 2 (e.g., the minority class in Figure 3(b) is uniformly distributed in the feature space), then no algorithm can perform better than random sampling. If the distribution of the majority class is not smooth and violates Assumption 1 (e.g., the majority class in Figure 3(c) consists of multiple narrow and sharp peaks just like the minority class), then the minority class cannot be identified with a clear clue. Based on the assumptions related to rare category examples, we propose the following definition of complex rare category analysis:

Definition 1.2. Complex Rare Category Analysis

Complex rare category analysis refers to the problem of detecting, characterizing, representing, and interpreting rare examples from underrepresented minority classes in a highly imbalanced dataset.

1.2 Challenges of Rare Category Analysis

We first discuss the key challenges associated with rare category analysis. Unlike the conventional anomalies, rare category examples have the following unique characteristics:

- **C1. Rarity.** It is often the case that the patterns of interest are sporadic in the collected data. For example, in financial fraud detection, the vast majority of financial transactions are legitimate; only a small number may be related to money laundering, and they often exhibit similar traits. Another example is network intrusion detection. New malicious network activities are hidden among huge volumes of routine network traffic, and network intrusions of the same type are often similar to one another.
- **C2. Label Scarcity.** Given the high skewness and non-separable nature of rare categories, labeling rare category examples is extremely expensive. In various real applications, such as money laundering detection, synthetic identity (ID) detection, and network intrusion detection, we are usually required to learn from a handful of labeled examples.
- **C3. Non-separability.** Rare categories are often non-separable from the majority classes in the feature space. For example, from the alarm of grid hacking (e.g., NotPetya) to the swift growth in cyber-criminals (e.g., coin-mining attacks), the past several years have served as a stark reminder of an emerging type of terrorist attack (i.e., *adversarial attacks*). Such malicious attacks are designed to bypass current detection systems by camouflaging themselves as normal instances, i.e., the support regions of normal instances, and malicious attacks are non-separable in the given feature space.
- **C4. Data Heterogeneity.** To identify the rare categories, we often need to collect raw data from different sources, at different time stamps using different techniques. Each data source might have different types of users and attributes. Of such highly heterogeneous raw data,

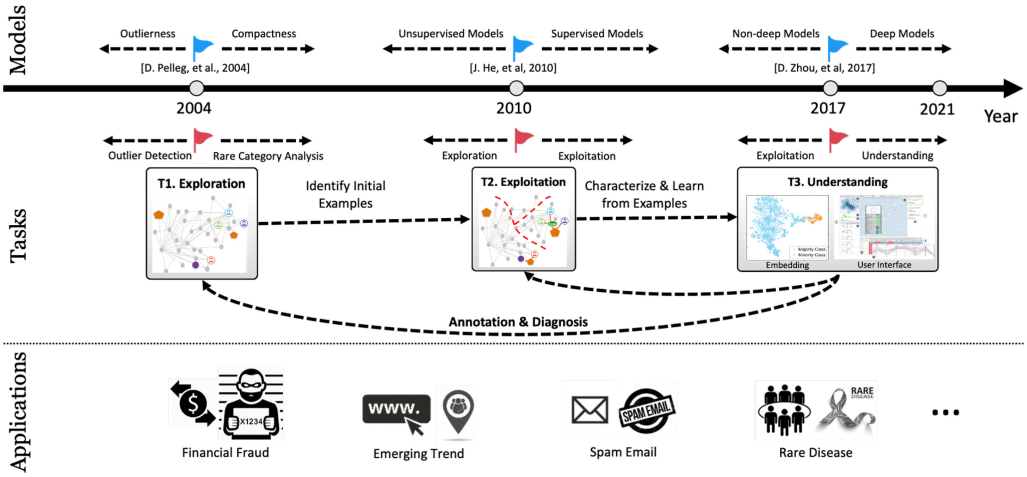


Fig. 4. Overview of complex rare category analysis.

only a subset (e.g., data sources, attributes) may be relevant to the identification of the rare category examples (e.g., security threats).

- **C5. Dynamics.** The raw input data continuously change over time; thus, extracting and modeling useful information from such dynamic systems plays an important role in rare category analysis.

1.3 An Overview of Complex Rare Category Analysis

Complex rare category analysis can be divided into four tasks: detecting, characterizing, representing, and interpreting rare category examples. Figure 4 presents an overview of rare category analysis, from the exploration step without any ground truth to the exposition step that aims to characterize rare examples with a compact representation, from representing rare patterns in a salient space to providing insightful interpretation over the prediction results.

- **T1: Rare Category Exploration.** In this task, we start from de-novo and do not have any label information for the data. The goal is to identify at least one example from each rare category with minimal help from labeling oracles. We begin with a comprehensive review of some relevant topics, including anomaly detection [7, 26], outlier detection [57], and predictive modeling on imbalanced domains [20]; then we elaborate on popular approaches to rare category exploration in both static and dynamic environments.
- **T2: Rare Category Exploitation.** In this task, we use one/a few labeled examples from each class that may have been obtained from the exploratory step discussed in T1. The primary goal is to capture a compact representation of the rare categories in various data formats by learning from a few-shot annotated data points. Depending on the availability of a multimodal representation of examples, rare category detection can be categorized as homogeneous rare category exploration or heterogeneous rare category exploration. In the former setting [52, 109], all the examples are homogeneous, as they are collected from the same data source or have identical traits; in the latter one [41, 79, 160, 163], we aim to identify rare categories in the presence of data heterogeneity.
- **T3: Understanding Rare Categories:** This task, which may entail representing rare examples in a salient embedding space, visualizing the data distribution, or interpreting the underlying prediction process in the previous steps (T1 and T2), serves as an investigation step for

end-users. We systematically discuss recent advances in representation learning [60, 166], interpretation [30, 84, 89], and visual analytic systems [81, 105] for rare category analysis.

1.4 Previous Work and Our Contributions to the Literature

Extensive literature surveys on the topics of imbalanced classification, anomaly/outlier detection, and fraud detection have been published in the past decade. In the context of imbalanced classification, He and Garcia [48] cover the development of research on learning from imbalanced data. In the context of anomaly/outlier detection, Hodge and Austin [57] introduces a literature review of contemporary techniques for outlier detection, and Chandola et al. [26] presents a comprehensive overview of the research on anomaly detection. A series of surveys and special-issue journal articles that target more specific problems and applications in the context of anomaly/outlier detection have also been published. In particular, Zimek et al. [173] studies the problem of unsupervised outlier detection in high-dimensional numerical data, Chandola et al. [27] considers anomalies in discrete sequences, Gupta et al. [45] focus on outlier detection for temporal data, Akoglu et al. [7] attends to graph-based anomaly detection algorithms, Chalapathy and Chawla [25] provide an overview of deep-learning-based anomaly detection techniques, and Edge and Falcone Sampaio [36] and Phua et al. [113] survey methodologies of fraud detection.

However, very few works present a comprehensive overview of complex rare category analysis. He [49] is one of the first to present an end-to-end investigation of rare categories, but the introduced methodologies are largely restricted to the tabular data in the static setting. Responding to the gap in the literature, here we provide a comprehensive and structured overview of complex rare category analysis from several perspectives:

- *Data Perspective.* Unlike the previous surveys on outlier detection, anomaly detection, and rare category analysis, our study reviews state-of-the-art techniques for handling rich data types, such as text/blogs, electronic health records, weblink graphs, and streaming data.
- *Task Perspective.* We provide a comprehensive overview of end-to-end complex rare category analysis, from rare category exploration without any label information, to rare category exposition characterizing rare examples with compact representation, from representing rare patterns in a salient embedding space to interpreting the prediction results and providing relevant clues for end-users' investigation.

1.5 Survey Roadmap

We present our survey in five parts sections. In Section 2, we focus on the de novo step of rare category analysis, i.e., rare category exploration, which aims to find the initial example from each minority class in the unsupervised setting. In Section 3, we discuss the following step, rare category exploitation, which relies on the output of rare category exploration (i.e., initially labeled examples) to characterize the rare categories. In Section 4, we stress the importance of interpretation in rare category analysis in various highly regulated industries, such as finance and healthcare. We systematically discuss how to understand rare category analysis from the following two aspects: (1) data diagnosis (i.e., *How are the data distributed? Which piece of information is most valuable for a given task?*); and (2) model diagnosis (i.e., *Why does the model make a certain prediction on a particular piece of information?*). Section 5 enumerates the popular benchmark datasets and representative baseline models. In Section 6, we conclude our study and share our thoughts regarding future research directions, such as how to ensure performance in the presence of noise, missing data, and adversarial examples and how to generate task-specific rare category examples (e.g., money laundering activity) given a specific domain (e.g., transaction networks). A general outline follows:

- **Section 2: Rare Category Exploration**
 - 2.1 Rare category exploration for static data
 - * 2.1.1 Tabular data
 - * 2.1.2 Graph-structured data
 - 2.2 Rare category exploration for dynamic data
 - * 2.2.1 Time-series data
 - * 2.2.2 Temporal graphs
- **Section 3: Rare Category Exploitation**
 - 3.1 Homogeneous rare category exploitation
 - * 3.1.1 Global approaches
 - * 3.1.2 Local approaches
 - 3.2 Heterogeneous rare category exploitation
 - * 3.2.1 Data heterogeneity
 - * 3.2.2 Task heterogeneity
- **Section 4: Understanding Rare Categories**
 - 4.1 Rare category representation
 - 4.2 Rare category interpretation
- **Section 5: Datasets and Representative Methods**
 - 5.1 Datasets
 - 5.2 Representative methods
- **Section 6: Conclusion and Future Directions**

2 RARE CATEGORY EXPLORATION

Considered the de novo step in rare category analysis, rare category exploration, a.k.a. rare category detection, aims to propose initial rare category examples to the labeling oracle in the given dataset. Given an unlabeled, imbalanced dataset, the most natural method of rare category exploration is random sampling. In this method we randomly draw examples without replacement to be labeled by the oracle until at least one example from each minority class has been identified. However, due to the rarity of rare categories (e.g., the proportion of a rare category is 0.01%), an extensive number (i.e., 10,000) of labeling requests is required to spot the rare example. It is thus worthwhile to develop more effective and efficient sampling strategies to address this rare category exploration problem. In this section, we formally define the problem of rare category exploration. We primarily aim to address (C1) rarity, (C4) data heterogeneity, and (C5) dynamics in rare category analysis.

PROBLEM 1. *Rare Category Exploration.*

Given: (i) an unlabeled dataset $\mathcal{D}$, (ii) a small budget $\mathcal{B}$ for querying domain experts.

Find: the set of initial examples Q from each rare category.

In real-world applications of rare category exploration, researchers have customized Problem 1 for different data formats, including tabular data, time-series data, and graph-structured data. In the following three subsections, we review the existing literature on rare category exploration from a data perspective (e.g., static data vs. dynamic data, tabular data vs. graph-structured data).

2.1 Rare Category Exploration for Static Patterns

The techniques currently used in the static setting have been primarily designed in active-learning-based schemes. The general procedure of this paradigm is presented in Figure 5. The key idea behind the sample-selection-based methods is to learn a detector that can gradually search for the potential rare examples by learning from the examples labeled by the oracle. We often assume that

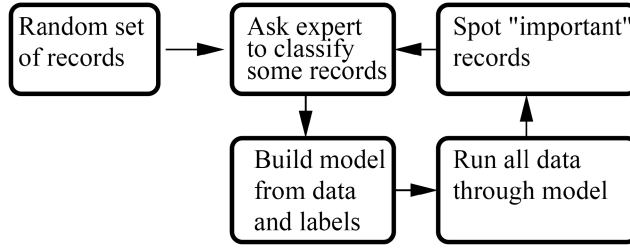


Fig. 5. A conceptual framework for rare category exploration [109].

the labels obtained from the oracles are under a fixed budget $\mathcal{B}$. Thus, the goal is to maximally improve the prediction accuracy with the minimum number of labeling requests to the oracle. As shown in Figure 5, rare category exploration proceeds iteratively. In each iteration, the program starts with the oracle annotating a few samples. Then the sample-selection-based methods are updated based on the newly labeled examples and further predict a small number of potential rare category examples in the sense that obtaining labels for them further improves the prediction accuracy. The program terminates once all the target rare categories have been captured. Next, we elaborate on the existing rare category exploration algorithms designed for tabular and graph-structured data.

2.1.1 Tabular Data. Given an unlabeled dataset with n samples $\mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$, where each sample $\mathbf{x}_i$, $i = 1, \dots, n$ comes with d -dimensional features $\mathbf{x}_i = \{x_i^1, \dots, x_i^d\}$, our goal is to identify at least one example from each class $y = 1, \dots, m$ with the minimum number of queries. The work of Peleg and Moore [109] is one of the first attempts at rare category exploration. The authors develop a mixture model to fit the data and design a family of hint selection methods to select the rare examples with the help of a human expert. Experimental results using different hint selection methods show the efficacy of the proposed rare category detection framework. He and Carbonell [52] study the rare category exploration problem when the minority classes are non-separable from the majority classes. Specifically, the authors develop a nearest-neighbor-based rare category detection algorithm named *NNDM*, which gradually selects examples with the maximum changes in the local density on a certain scale and asks for labeling from the oracle. The algorithm works as follows: Given the prior p of each rare category C , the algorithm first estimates the number of examples $K = n \times p$ in C . Then, the algorithm represents each example $\mathbf{x}_i$ in terms of a vector that is composed of the distances from its K -nearest neighbors. By obtaining the minimum distance r among all samples, the algorithm constructs a hyper-ball with radius r for each example (as center) and computes the local density as the number of enclosed examples within the hyper-ball:

$$\text{Hyberball}(\mathbf{x}_i, r) = \{\mathbf{x} | \mathbf{x} \in \mathcal{D}, \|\mathbf{x} - \mathbf{x}_i\| \leq r\} \quad (1)$$

$$\text{LocalDensity}(\mathbf{x}_i) = |\text{Hyberball}(\mathbf{x}_i, r)| \quad (2)$$

To measure the degree of local density changes around each sample (i.e., hyper-ball), the authors propose using the difference in the local density between $\mathbf{x}_i$ and its neighbors:

$$\begin{aligned} &\text{DensityChange}(\mathbf{x}_i) \\ &= \begin{cases} \max_{\mathbf{x}_k \in \text{Hyberball}(\mathbf{x}_i, r)} [\text{LocalDensity}(\mathbf{x}_i) - \text{LocalDensity}(\mathbf{x}_k)] & \mathbf{x}_i \text{ is labeled} \\ -\infty & \text{Otherwise} \end{cases} \end{aligned} \quad (3)$$

In particular, the function *DensityChange* assigns each labeled example's scores with $-\infty$ to avoid duplicated labeling requests. Finally, the algorithm returns the examples with the largest local

density changes to the oracle for labeling:

$$\text{Query} = \operatorname{argmax}_{\mathbf{x}_i \in \mathcal{D}} \text{DensityChange}(\mathbf{x}_i) \quad (4)$$

Theoretical analysis shows that the methods effectively select examples on both the boundary and in the interior of the rare categories. In particular, when the rare categories are compact, the majority class distribution is locally smooth, and the sample space is sufficiently large, NNDM is guaranteed to identify at least one example coming from each minority class with probability $1 - \delta$ after $\lceil \frac{2\alpha}{r_2} \rceil$ iterations, where $\delta \in (0, 1)$, α , and r_2 are all positive parameters.

Despite the promising results of NNDM with theoretical guarantees, the algorithm's performance largely relies on prior information (e.g., number of classes, proportion of minority classes). To alleviate the restriction on prior knowledge, He and Carbonell [50] propose a prior-free rare category detection algorithm named *SEDER*. The intuition of this algorithm lies in the observation that a region with an abrupt local density change has a higher probability of observing the rare classes. Unlike He and Carbonell [52] and He et al. [53], *SEDER* picks the potential rare examples with large neighborhood density changes for labeling by performing semi-parametric density estimation. *SEDER* introduces a prior-free *DensityChange* function, which is defined as the norm of the gradient of the local density function:

$$\text{DensityChange}(\mathbf{x}_i) = \sqrt{\sum_{l=1}^d \frac{(\sum_{k=1}^n \text{DensityEstimation}(\mathbf{x}_i)(x_i^l - b^l x_k^l))^2}{((\sigma^l)^2 b^l)^2}} \quad (5)$$

where x_i^l denotes the l th feature of $\mathbf{x}_i$, σ^l denotes the bandwidth of the l th feature, *DensityEstimation* is a kernel density estimation function with Gaussian kernel [37], and b^l is a positive parameter. In the presence of noisy data and irrelevant features, He and Carbonell [51] formulate the rare category exploration problem via a co-selection scheme, which recovers the relevant features and representative examples from the rare categories. To obtain the optimal sets of relevant features and rare examples, He and Carbonell propose an effective searching procedure (i.e., *PALM*) based on an augmented Lagrangian function to solve the optimization problem. In particular, *PALM* is designed in an alternative fashion so that the relevant features and the minority class examples can be found.

Recently, a paradigm shift has been observed in the data mining community: from how to accurately detect rare categories to how to ensure trustworthiness in rare category exploration. For instance, Ahmadian and Lindsten [4], Kaur et al. [67], and Liu et al. [83] study how to ensure robustness in detecting rare events, especially in the out-of-distribution setting. Shekhar et al. [120] point out that algorithmic fairness and rare category exploration are dual problems when encountering the data imbalance issue. The authors propose a fairness-aware method that jointly optimizes prediction accuracy and statistical parity. Deng et al. [32] study the hardness bias in the autoencoder-based rare category detection methods and propose a plug-and-play error calibration method to mitigate the hardness bias issue.

2.1.2 Graph-Structured Data. In many fields, graphs offer a unifying data structure for modeling relational data. As a result, extensive research related to rare category exploration has been conducted to spot the rare category entities in graph-structured data. Formally speaking, given an unlabeled graph $\mathcal{D} = \mathcal{G}(\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ and $\mathcal{E}$ denote the sets of nodes and edges in $\mathcal{G}$, respectively, our target is to identify initial nodes and edges from each of the rare categories. He et al. [53] extends the idea of He and Carbonell [52] to graph-structured data by proposing a graph-based rare category detection algorithm named *GRADE*. The algorithm starts computing a global similarity matrix $\mathbf{S}$ motivated by manifold ranking [169], which is used to obtain a compact representation

of the examples from the minority classes:

$$\mathbf{S} = (\mathbf{I} - \alpha \mathbf{D}^{-\frac{1}{2}} \mathbf{A} \mathbf{D}^{-\frac{1}{2}})^{-1} \quad (6)$$

where $\mathbf{I}$ denotes the identify matrix, $\mathbf{D}$ denotes the diagonal matrix, and $\mathbf{A}$ denotes the adjacency matrix of $\mathcal{G}$. A prior-guided k -nearest-neighbor matrix is then computed to capture the sharp local density changes near the boundary of minority classes and thus make it easier to capture the rare patterns. In addition to developing *GRADE*, the authors develop a variation named *GRADE-LI*, which requires only an upper bound on the proportion of each rare category. *GRADE-LI* can work with the data when a detailed class-membership distribution of the data is not available to the users.

Except for the plain graphs, data often exhibit node-level and edge-level heterogeneity for various critical tasks in security, finance, and medicine, among other fields. Each node or edge is associated with a specific type in such data (referred to as a *heterogeneous graph*). To accommodate heterogeneous graphs, researchers have studied how to detect anomalies and rare events in the unsupervised setting. For example, Sun et al. [127] propose the notion of neighborhood formation for a bipartite graph, which computes the relevance scores of all nodes to a query node v and define the neighborhood of v as the set of nodes with higher relevance scores. Based on the neighborhood formation, the authors develop an anomaly detection algorithm to spot the abnormal nodes with low “normality” scores. Based on similar intuition, Tong and Lin [132] propose a non-negative residual matrix factorization framework named *NrMF*, which aims to detect the malicious group of entities as well as provide interpretation of prediction results for data analysts. *NrMF* is built upon the conventional matrix factorization mechanism,

$$\mathbf{A} = \mathbf{F}\mathbf{G} + \mathbf{R} \quad (7)$$

where $\mathbf{F}$ and $\mathbf{G}$ are low-rank factorized matrices, and $\mathbf{R}$ is the residual matrix that indicates rare examples on the bipartite graphs. To interpret the predictions via *NrMF*, Tong and Lin impose a non-negative constraint on the residual matrix $\mathbf{R}$, such that all entries in $\mathbf{R}$ are larger than 0. Intuitively, higher value of the entry in $\mathbf{R}$ indicates a higher probability of existing rare and abnormal examples. To solve the optimization problem of non-negative residual matrix factorization, the authors develop a fast optimization algorithm to incrementally compute the residual matrix via the rank-1 approximation, which runs in linear w.r.t. the size of the graph.

More recently, extensive attempts have been made to predict rare category events by developing customized **graph neural networks (GNNs)** to depict the structure of a graph and learn representations of various graph signals (e.g., nodes, edges, subgraphs). GNNs-based approaches are generally designed based on the message-passing scheme, which is given as follows:

$$\mathbf{h}_v^{(l+1)} = \text{UPDATE}^{(l)}(\mathbf{h}_v^{(l)}, \text{AGGREGATE}^{(l)}(\mathbf{h}_u^{(l)}, \forall u \in \mathcal{N}(v))) \quad (8)$$

$$= \text{UPDATE}^{(l)}(\mathbf{h}_v^{(l)}, \mathbf{m}_{\mathcal{N}(v)}^{(l)}) \quad (9)$$

where $\mathbf{h}_v^{(l+1)}$ denotes the embedding of node $v \in \mathcal{V}$ at the $(l+1)$ th-layer, $\text{UPDATE}^{(l)}$ and $\text{AGGREGATE}^{(l)}$ are both differentiable functions for the l th-layer, and $\mathbf{m}_{\mathcal{N}(v)}^{(l)}$ is the aggregated “message” from node v ’s neighborhood $\mathcal{N}(v)$. Deng et al. [34] propose a **graph convolutional network (GCN)**-based framework for predicting emerging events that requires capturing contextual information from the raw data. The proposed framework first extracts graph representations of the events; it then documents and learns to predict the occurrence of future events and to identify the events of interest (i.e., rare categories). Zhong et al. [162] studies the problem of rare category detection in videos, where each video is decomposed into n snippets (i.e., nodes in the constructed video graph), the edges between each pair of snippets are defined based on feature similarity, and

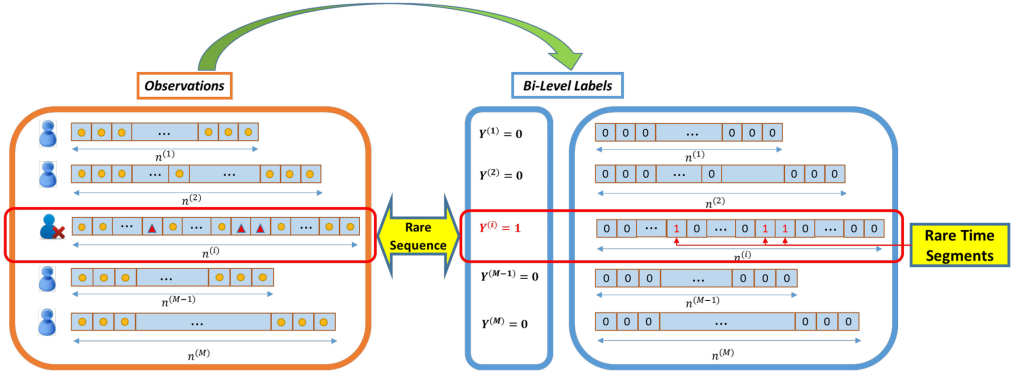


Fig. 6. An illustration of the bi-level rare categories (i.e., rare sequences and rare time segments) in a time-series database [164].

the rare events are represented as abnormal actions. To solve this problem, a GCN is built to simultaneously clean noisy signals in the constructed video graph and predict abnormal actions. Goodge et al. [42] investigate the relationship between the local outlier methods [69] and the message-passing framework, which motivates them to develop a GNN-based method to identify local rare examples.

2.2 Rare Category Exploration for Dynamic Patterns

The existing literature on detecting rare patterns in temporal data is very rich, with no universal abstract categorization capable of capturing all the developed techniques in this direction. Accordingly, we here focus on rare category exploration for time-series data and temporal graphs.

2.2.1 Time-Series Data. In the setting of time-series data, the rare categories can be observed at both the sequence level and the segment level, as shown in Figure 6. For example, most electrocardiogram (ECG) signals are normal (i.e., collected from healthy people); only a small portion of them is abnormal (i.e., collected from patients with heart disease). Moreover, within the abnormal ECG signals, only a few segments are abnormal (e.g., heart arrhythmia); the rest are normal. Early studies of rare category exploration for time-series data [18, 21, 22, 65, 80, 97, 100, 108, 118, 150] are closely related to the outlier detection and disorder detection methods. They rely largely on distanced-based mechanisms [97, 101, 116, 119] that define various similarity measurements [45] of temporal sequences and segments and then identify rare patterns deviating from the normal ones. For instance, Cao et al. [22] propose a scalable distance-based detection algorithm for high-volume data streams. This algorithm has been found to be optimal in terms of CPU costs. Begum and Keogh [18] study the problem of discovering rare time-series motifs (i.e., repeated subsequences) from unbounded streams. To address the rarity issue of the time-series motif in a never-ending stream, Begum and Keogh develop a “sticky cash” algorithm that adopts a Bloom filter to remember every incoming subsequence and efficiently detects rare motifs in the unbounded real-valued time series. To facilitate the use of distance-based methods, Yeh et al. [150] introduce a fast computational algorithm for time-series/subsequence all-pairs-similarity-search that has strong implications for and promising results for the task of time-series discord discovery. Finally, Goodge et al. [43] introduce a robust random-cut data structure to produce a sketch or synopsis of time-series data. Using this structure, the authors propose a scalable anomaly detection algorithm by gradually updating the time-series sketch in a continuous data stream.

A key motivation of the aforementioned methods is that the distributions of rare categories (minorities) deviate from the normal distribution (majorities). However, there are some obvious limitations to this idea in practice. The identified examples are often not the targets of researchers' interest, which are drawn from noise or combinations of irrelevant features. Zhou et al. [164] assume that the targets of our interest are rare and drawn from a compact distribution. Under this assumption, Zhou et al. propose a bi-level generative model that aims to jointly characterize the rare temporal pattern at the time-series and subsequence levels. Recent advances in deep learning architectures have led to a surge in research interest related to developing deep models to characterize and detect rare temporal patterns in high-impact applications. Tuor et al. [134] present a deep unsupervised framework for detecting insider threat in online data streams. This framework outputs a ranked list of anomaly scores of individual user behaviors; Li et al. [77] develop a **generative adversarial network (GAN)** for unsupervised multivariate anomaly detection. Unlike conventional distance-based methods and supervised methods, the proposed framework detects rare temporal patterns by using the GAN-trained generator and discriminator to compute the **discrimination and reconstruction anomaly score (DR-Score)**; Shen et al. [121] propose a recurrent network ensemble called the *recurrent autoencoder*, which is designed to characterize and capture the abnormal time-series segments at multiple resolutions; finally, Deng and Hooi [33] and Zhang et al. [159] propose modeling the between-sensor dependence relationship to identify abnormal time segments in multivariate time series.

2.2.2 Temporal Graphs. Many real-world systems, including social networks, communication networks, and gene interaction networks, are intrinsically dynamic and can be represented as temporal graphs. In the past few years, researchers have proposed several rare category exploration models for temporal graphs [3, 13, 34, 59, 76, 89, 90, 115, 125, 140, 149, 153, 167, 168]. The existing work can be categorized as discrete temporal graphs [154] or continuous temporal graphs [85, 103, 171] based on the method of data collection used in the particular application domain.

The discrete temporal graphs are often referred to as a time-evolving graphs wherein the data $\mathcal{D}$ is presented as a sequence of snapshots $\hat{\mathcal{G}} = \{\mathcal{G}^{(1)}, \mathcal{G}^{(2)}, \dots, \mathcal{G}^{(T)}\}$. Each snapshot $\mathcal{G}^{(t)} = (\mathcal{V}^{(t)}, \mathcal{E}^{(t)})$, $t = 1, 2, \dots, T$ is composed of a collection of nodes $\mathcal{V}^{(t)}$ and edges $\mathcal{E}^{(t)}$ at timestamp t . To identify rare examples on $\hat{\mathcal{G}}$, the static methods are extended to the dynamic setting. For example, Sun et al. [126] propose a parameter-free model that can monitor grouped outliers and their changes in a stream of graphs. The algorithm is designed based on **minimum description length (MDL)**, which allows users to discover the changes in communities as well as in points in time. Araujo et al. [14] develop a fast incremental tensor analysis approach that can discover both transient and periodic/repeating communities in dynamic graphs. Sricharan and Das [125] define a commute-time distance that captures node relationship changes and allows traditional distance-based methods to be used with discrete temporal graphs. Lee et al. [76] propose a discrete-time exponential-family random graph model to identify clusters in time-evolving graphs. Manzoor et al. [90] study the problem of anomaly detection in streaming heterogeneous graphs by proposing a clustering-based anomaly detection approach that can simultaneously address the heterogeneity and streaming nature of the input data. In particular, Manzoor et al. introduce a novel embedding mechanism that can encode the heterogeneous streaming graph into a vector representation that is then used to perform clustering and identify anomalous patterns. Appel et al. [13] proposes a factorization framework that can jointly model the distribution of dynamic connections and attributes and track the evolution of evolving communities. Despite their success, the detection algorithms often suffer from high computational costs, especially when extensive snapshots are given or when the algorithms are applied in the online setting. To address this issue, Zhou

et al. [167, 168] propose an incremental rare category exploration scheme that aims to gradually update the static rare category exploration models based on the local changes on a new snapshot without learning from scratch. In particular, Zhou et al. propose a closed-form solution to update the global similarity matrix $\mathbf{S}$ (defined in Equation (6)) in the dynamic setting:

$$\mathbf{S}^{(t)} = \mathbf{S}^{(t-1)} + \alpha \frac{\mathbf{S}^{(t-1)} \mathbf{u} \mathbf{v}^T \mathbf{S}^{(t-1)}}{\mathbf{I} + \mathbf{v}^T \mathbf{S}^{(t-1)} \mathbf{u}} \quad (10)$$

where $\mathbf{u}$ and $\mathbf{v}$ are one-hot vectors indicating the source nodes and target nodes of the updated edges, respectively; $\mathbf{I}$ denotes the identity matrix; and $\mathbf{S}^{(t)}$ and $\mathbf{S}^{(t-1)}$ are the global similarity matrices at the timestamps t and $(t-1)$, respectively. Moreover, to efficiently and effectively compute the density changes, $\text{DensityChange}(\mathbf{x}_i)$, a theoretical condition, is provided to determine whether the hyper-ball $\text{Hyberball}(\mathbf{x}_i, r)$ is required to update from the previous timestamp. Yasami and Safaei [149] study the problem of anomaly detection in dynamic social networks, wherein both network structure and node attributes are observed over time. The proposed framework jointly models two processes, i.e., (1) the normal modeling component and (2) the anomaly detection component, to track the abnormal relationship between nodes' features and link generation in dynamic social networks. Agarwal et al. [3] introduce a novel community scoring metric called *permanence* and proposes an incremental algorithm to track the evolution of network communities in the dynamic setting. The theoretical analysis shows that the updating procedure of the proposed algorithms leads to permanence maximization in the dynamic networks. Furthermore, to guarantee model efficiency and algorithmic fairness, Fu et al. [39] introduce a fairness-aware clique-preserving spectral clustering algorithm that generalizes the static clique clustering methods to the dynamic setting via fairness-aware edge filtering and incremental eigenpair updating.

Continuous temporal graphs are also called *fine-grained temporal graphs* or *temporal interaction graphs*, where the temporal graph is presented as a sequence of timestamped edges. It is intractable to directly generalize the static rare category exploration approaches used in discrete temporal graphs to continuous temporal graphs. For this reason, Manzoor et al. [90] propose to represent continuous temporal graphs with a vector representation, which is easy to compute and preserves the contextual information of the continuous temporal graphs. With the learned continuous temporal graph representation, Manzoor et al. develop a fast and memory-efficient detection algorithm to process any incoming nodes and edges and identify anomalies in real time. Hollocou et al. [59] subsequently studies the problem of identifying grouped anomalies in the edge streaming setting. In particular, the data are presented as a sequence of streaming edges. The authors propose a streaming algorithm with the theoretical justification that performs graph clustering with only three integers per node and does not keep any edge in memory. Amjadi and Tulabandhula [10] propose a block-structured time-series model for detecting communities on time-evolving graphs. This approach can capture both the link persistence and community persistence over time.

3 RARE CATEGORY EXPLOITATION

In the previous section, we introduced the current literature on rare category exploration, a problem that aims to identify at least one example from each minority class. A follow-up to rare category exploration, rare category exploitation aims to learn from the identified examples and capture all the rare examples in a given dataset. In Figure 7, we present an example of rare category exploitation on a synthetic dataset with one majority class and four minority classes. The key observations are as follows: (1) the distribution of the majority class is smooth and (2) the support regions of the four minority classes are compact. That is, instances in each minority class are self-similar and form a compact representation. In the past decades, the question of how to

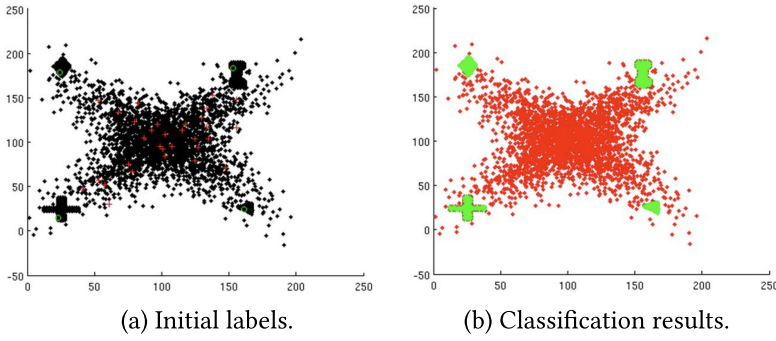


Fig. 7. An example of rare category exploitation with one majority class and four minority classes [54]. The labeled minority class examples are shown in green, the labeled majority class examples are shown in red, and the unlabeled examples are shown in black.

explore the compactness of rare categories has attracted considerable interest from the data mining and machine learning communities. In this section, we delve into the problem of rare category exploitation under two scenarios: (1) homogeneous data and (2) heterogeneous data. Whereas in the former setting, data are collected from the same source and represented in the same format in the latter setting, we aim to capture rare examples in the presence of data and task heterogeneity.

3.1 Homogeneous Rare Category Exploitation

Formally, we let $\mathcal{X}$ and $\mathcal{Y}$ denote the sample space and label space, respectively. We are given a dataset $\mathcal{D} = \{\mathcal{D}_l, \mathcal{D}_u\}$, where $\mathcal{D}_l = \{(\mathbf{x}_i)\}_{i=1}^{N_l}$ consists of N_l obtained annotated samples, $\mathcal{D}_u = \{(\mathbf{x}_j)\}_{j=1}^{N_u}$ denotes N_u unlabeled examples, and $\mathbf{x}_i, \mathbf{x}_j \in \mathcal{X}$, $y_i \in \mathcal{Y}$. It is often the case that, due to the (C1) rarity and label (C2) scarcity challenges, annotated data N_l are scarce. In rare category exploitation, our goal is to produce a prediction function $f(\cdot) : \mathcal{X} \rightarrow \mathcal{Y}$ by learning from $\mathcal{D}_l$ and to accurately characterize unknown rare category examples in $\mathcal{D}_u$. Given this, we present the following definition of rare category exploitation in the homogeneous setting. In this problem setting, we mainly study (C1) rarity, (C2) label scarcity, (C3) non-separability, and (C5) dynamics in RCA.

PROBLEM 2. *Homogeneous Rare Category Exploitation.*

Given: a dataset $\mathcal{D} = \{\mathcal{D}_l, \mathcal{D}_u\}$ with scarce labeled examples $\mathcal{D}_l = \{(\mathbf{x}_i)\}_{i=1}^{N_l}$ and extensive unlabeled examples $\mathcal{D}_u = \{(\mathbf{x}_j)\}_{j=1}^{N_u}$.

Find: (i) a prediction function $f(\cdot)$ to characterize rare category examples, (ii) a set of unlabeled examples that are likely coming from the rare categories.

We categorize the current approaches to homogeneous rare category exploitation as either global or local. The first category aims to capture the global data distribution for characterizing rare categories, whereas the second category focuses on exploring rare categories within a single or a few local regions (e.g., the neighborhood of the identified rare category examples).

3.1.1 Global Approaches. This category of approaches exploits both the labeled and unlabeled examples to learn the class-membership distribution. Essentially, these methods transform the rare category exploitation problem into a classification problem with highly skewed data distribution. However, unlike the well-studied imbalanced classification problem that aims to maximize

the overall classification accuracy (majority classes and minority classes), rare category exploitation emphasizes learning and characterizing the minority classes. Early rare category exploitation work addresses the (C1) rarity and (C3) non-separability challenges. One representative approach is hyper-ball-based rare category exploitation [54, 55]. The key idea behind approaches is to formulate rare category exploitation as an optimization problem: minimizing the radius R of the hyper-ball that well encloses the rare categories. The general objective function for the binary case (i.e., one majority class and one minority class) can be written as follows.

$$\begin{aligned} \min_{R, \mathbf{c}} \quad & R^2 \\ \text{s.t.}, \quad & \|\mathbf{x}_i - \mathbf{c}\|^2 \leq R^2, \mathbf{x}_i \in \mathcal{D}_l^{\text{maj}} \\ & \|\mathbf{x}_j - \mathbf{c}\|^2 \geq R^2, \mathbf{x}_j \in \mathcal{D}_l^{\text{min}} \\ & \|\mathbf{x}_k - \mathbf{c}\|^2 \geq R^2, \mathbf{x}_k \in \mathcal{D}_u \end{aligned} \quad (11)$$

where R denotes the radius of the hyper-ball, $\mathbf{c}$ denotes the center of the hyper-ball, $\mathcal{D}_l^{\text{maj}}$ denotes the set of labeled majority class examples, and $\mathcal{D}_l^{\text{min}}$ denotes the set of labeled minority class examples. Moreover, to ensure the compactness of the hyper-ball, the authors introduce three constraints on the optimization problem: (1) the labeled examples from the majority classes should be outside of the hyper-ball; (2) the labeled examples from the minority class should be inside of the hyper-ball; and (3) the hyper-ball should enclose as many unlabeled examples as possible. To tackle the non-separability challenges, RACH [54] provides a relaxed solution of Equation (11) by introducing slack variables to allow the miss-classified instances and then converting the problem into a convex optimization problem. The relaxed problem can be easily solved in its dual form via the projected subgradient method. Following He et al. [54], He et al. [55] introduces a kernelized rare category exploitation algorithm that, generalizes RACH to model complex shapes of the target rare categories' support regions by projecting to the high-dimensional feature space induced by kernels. The kernel method provides more flexibility when characterizing complex rare categories in real-world applications.

Several more recent studies consider the problem of rare category exploitation in the presence of (C2) scarce and noisy labels. Haque et al. [46], study the problem of novel class classification on data streams. Instead of relying on redundant human-annotated labels or prior knowledge, they propose a semi-supervised framework to track the confidence changes of classifiers in order to detect the novel concept drifts and thus identify the novel classes. Zhou et al. [166] propose a bi-level learning mechanism (shown in Figure 8) in which, a teacher model (i.e., curriculum learning scheme) gradually augments the training set with pseudo labels, and the student model (i.e., rare category characterization) returns the prediction results together with the prediction confidence to the teacher model. In general, the teacher model and the student model are trained in a mutually beneficial way, enabling the model to achieve better prediction accuracy in both rarity and label scarcity. Any real-world applications (e.g., video classification [162]), in addition featuring label scarcity, come with complex data and noisy labels, which require machine learning models to be trained with only weak supervision [172]. Zhong et al. [162] propose a GCN to clean the noisy labels. The GCN-based label noise cleaner is designed to provide supervisory signals from high-confidence snippets to low-confidence snippets, such that the classifier can be trained with "clean" supervision. Xi et al. [144] study the problem of online transaction fraud detection. To handle the noisy and complex user behavior information, they develop the **dual importance-aware factorization machine (DIFM)** that captures field value variations and field interactions simultaneously to support online transaction fraud detection. Lin et al. [82] propose a multi-resolution rare category exploitation approach to identify instances of online credit payment frauds at different

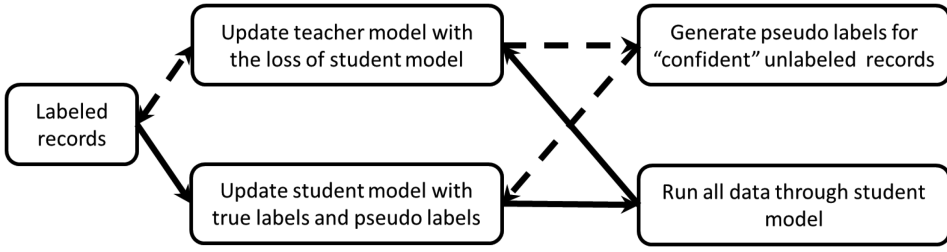


Fig. 8. A conceptual framework of bi-level learning for rare category exploitation [166] that is composed of a student learning model and a teacher learning model. The dashed arrows indicate the learning process of the teacher model that aims to generate pseudo labels; the solid arrows indicate the learning process of the student model that is designed to predict rare category examples.

granularities. Wu et al. [143] study rare category exploitation under various data contamination, by proposing an information-theoretic bound of performance degradation in terms of the data contamination ratio.

3.1.2 Local Approaches. The local algorithms are also referred to as seeded or targeted algorithms [56, 112, 151, 170]. This category of approaches essentially tackles rare category exploitation as a local clustering problem. The key idea is to treat the labeled examples as seeds and return a compact cluster near the seed examples without exploring the whole dataset. Azmandian et al. [16] present a local kernel density ratio feature selection framework that seeks a salient feature subspace in which the normal data points form a high-density region and the rare examples form a low-density region. In the learned subspace, the rare examples stand out and are easily distinguished by detection algorithms. Spielman and Teng [124], present a design of a local clustering algorithm for massive graphs. The local algorithm is able to find a compact local cluster near a seed example and the running time is near-linear with respect to the size of the returned cluster. Andersen et al. [11], derive a mixing result for PageRank and propose a fast approximation method to compute PageRank vectors. Using this method, the authors develop an improved local graph partitioning algorithm that enables polylogarithmic time complexity with respect to the number of edges in the given graph. Pang et al. [107] propose a local and interactive rare event detection algorithm, that leverages the feedback from an anomaly-biased simulation environment and continuously updates the learned abnormality to novel rare categories.

Although the conventional local clustering algorithms are primarily designed for low-order connectivity patterns (e.g., edge in Table 1), many applications exhibit high-order connectivity patterns (e.g., star in Table 1). For example, the loop-shaped structure plays a significant role in detecting communities in user-item networks [170]; a cluster of star-shaped structures often serves a red flag indicating potential instances of synthetic ID fraud in the **personal identifiable information (PII)** networks [102]. To account for the crucial high-order connectivity patterns in real-world systems, Zhou et al. [170] and Yin et al. [152] offer innovative studies of the problem of local high-order graph clustering. In particular, the authors introduce the notion of high-order conductance and the high-order diffusion kernel with the high-order Markov chain as the underlying model, which allows end-users to model various types of high-order structures. Moreover, they generalize the well-known local clustering algorithms [11, 124] to the high-order setting and provide a theoretical analysis of the effectiveness and scalability of the proposed algorithms. Many recent studies study the high-order connectivity patterns in more complicated scenarios, such as modeling evolving high-order structures on fine-grained temporal networks [165], capturing

Table 1. Illustrative Examples of High-Order Connectivity Patterns and Local Clustering Algorithms

Connectivity Pattern $\mathbb{N}$	Illustration	Order of $\mathbb{N}$	Clustering Algorithms	Applications
Vertex		1 st -order	N/A	N/A
Edge		2 nd -order	1 st -order	Rare category detection [53]
3-node Line		3 rd -order	2 nd -order	Social community detection [71]
Triangle				
k -node Star		k^{th} -order	$(k - 1)^{\text{th}}$ -order	Synthetic identity detection [102]

higher-order clusters in heterogeneous networks [23], and tracking high-order structures on time-evolving graphs [38].

3.2 Heterogeneous Rare Category Exploitation

In the era of big data, many application domains exhibit multiple types of heterogeneity, including data heterogeneity that originates from multiple information sources and task heterogeneity that originates from multiple application domains. In this subsection, we organize the existing work in the context of data heterogeneity and task heterogeneity.

3.2.1 Data heterogeneity. Learning from data heterogeneity is often referred to as multi-view learning, as the data are collected from different sources, enabling multiple views. In the data heterogeneity setting, rare category exploitation becomes even more challenging than in the data homogeneity setting due to the existence of multiple views. It is crucial to leverage multiple views and identify the relevant features for distinguishing between the majority and minority classes. With the aim of addressing (C1) rarity, (C2) label scarcity, and (C4) data heterogeneity in rare category analysis, we formally define the problem as follows:

PROBLEM 3. Multi-View Rare Category Exploitation.

Given: (i) a dataset $\mathcal{D}$ collected from V views; (ii) a small set of labeled examples $\mathcal{D}_l = \{(\mathbf{x}_i)\}_{i=1}^{N_l}$.
Find: (i) a prediction function $f(\cdot)$ to characterize rare category examples, (ii) a set of unlabeled examples that are likely coming from the rare categories.

A pivotal step in addressing Problem 5 is to leverage view consistency and learn a unified representation to improve the performance of rare category exploitation. In the past decades, researchers have proposed extensive multi-view learning frameworks [9, 40, 63, 64, 78, 79, 79, 91, 122, 157, 160] for anomaly detection that could potentially solve Problem 5. However, these methods are mostly unsupervised and/or based on some heuristic functions, which means the methods might not well capture the characteristics of rare categories. *MUVIR* [163] is one of the principled efforts that has been proposed to address Problem 5. *MUVIR* provides a generic rare category detection solution that is able to integrate the existing single-view rare category exploitation models in order to compute the overall posterior probability of each example. The key idea is to exploit the relationship among multiple views and estimate the overall posterior probability of examples coming from rare categories given data from multiple views. In particular, given data with multi-view features, one can train V distinct rare category exploitation models upon V views and compute view-specific posterior probability $P(y = \text{rare category} | \mathbf{x}^v)$ with respect to the v^{th} view. The authors propose the following theorem to effectively integrate view-specific posterior probability $P(y = \text{rare category} | \mathbf{x}^v)$ in a model-agnostic way:

THEOREM 3.1 ([163]). *If the features from multiple views have weak dependence given the class label y_i [1], i.e., $P(\mathbf{x}|y = \text{rare category}) \geq \alpha \prod_{v=1}^V P(\mathbf{x}^v|y = \text{rare category})$, $\alpha > 0$, then*

$$P(y = \text{rare category}|\mathbf{x}) \geq C \left(\prod_{v=1}^V P(y = \text{rare category}|\mathbf{x}^v) \right) \times \left(\frac{\prod_{v=1}^V P(\mathbf{x}^v)}{P(\mathbf{x})} \right) \quad (12)$$

where $C = \frac{\alpha}{(p^2)^{V-1}}$ is a constant.

Using Theorem 3.1, the overall posterior probability $P(y = \text{rare category}|\mathbf{x})$ can be approximated as follows:

$$P(y = \text{rare category}|\mathbf{x}) = \prod_{v=1}^V P(y = \text{rare category}|\mathbf{x}^v) \left(\frac{\prod_{v=1}^V P(\mathbf{x}^v)}{P(\mathbf{x})} \right)^d \quad (13)$$

where the marginal probabilities $P(\mathbf{x})$ and $P(\mathbf{x}^v)$ can be estimated via kernel density, and d is a non-negative parameter that balances the importance of the term related to the marginal probabilities. A modified version of *MUVIR* has been proposed for handling problems when the exact priors of minority classes are unknown. In the past few years, researchers have explored many high-impact applications of rare category exploitation that can be formulated as multi-view rare category exploitation problems. Among them, Lee et al. [75] propose an attention-based visual question-answering network to jointly model the input images and corpora and identify system errors or abnormal events; Chen et al. [29] study the problem of multi-modal video anomaly detection, proposing a bi-directional predictive network to regularize the prediction task from pixel-wise, cross-modal, and temporal-sequence levels; and Shen et al. [123] propose an anomaly segmentation network for localizing defective areas in large-scale industrial manufacturing environments.

Another group of approaches to multi-view rare category exploitation consist of the semi-supervised classification models. For example, Nie et al. [104] propose a multi-view framework with adaptive neighborhood learning. The framework integrates multi-view clustering and semi-supervised classification, which allows learning attention for each view automatically. Zhang et al. [156] develop a multi-view rare category exploitation framework for Alzheimer's disease diagnosis. They propose a multi-layer multi-view framework that automatically constructs a shared latent representation across multiple views and learns the mapping functions from the multi-view features to the prediction labels. Wang and Lan [138] assume the multiple views of the majority class examples are drawn from a unique distribution with different projection functions. Given this assumption, the authors propose a hierarchical Bayesian model that computes the outlieriness of unlabeled samples and thus identifies rare examples.

3.2.2 Task heterogeneity. This subsection provides an overview of the rare category exploitation algorithms that have been proposed in the context of task heterogeneity. Formally speaking, in the presence of multiple tasks, we are given a dataset $\mathcal{D} = \{\mathcal{D}^1 \cup \dots \cup \mathcal{D}^S\}$ that comes from S applications domains, and a small set of labeled examples from each task $\mathcal{D}_l^s = \{(\mathbf{x}_i^s)\}_{i=1}^{N_l^s}$, where N_l^s denotes the number of labeled examples in $\mathcal{D}^s$. Our goal is to capture the multimodalities from different tasks and domains and learn a prediction model to accurately characterize the rare categories from different tasks and domains. The problem, which aims to address (C1) rarity, (C2) label scarcity, (C3) non-separability, and (C5) dynamics in rare category analysis, is commonly defined as follows:

PROBLEM 4. Multi-task Rare Category Exploitation.

Given: (i) a dataset $\mathcal{D} = \{\mathcal{D}^1 \cup \dots \cup \mathcal{D}^S\}$, where $\mathcal{D}^s$ is the dataset for the s th task, $s = 1, \dots, S$; (ii) a small set of labeled examples from each task $\mathcal{D}_l^s = \{(\mathbf{x}_i^s)\}_{i=1}^{N_l^s}$.

Find: (i) a prediction function $f(\cdot)$ to characterize rare category examples in each task $\mathcal{D}^s$, (ii) a set of unlabeled examples that are likely coming from the rare categories in each task $\mathcal{D}^s$.

The past decades have produced much literature studying Problem 4. This literature can be categorized in one of two ways based on its objectives. The first category aims to learn a mixture of prediction functions that maximize the overall accuracy across all tasks. For example, Yang et al. [146] study the problem of rare category exploitation in the presence of multiple tasks and multiple views by proposing a joint optimization framework called M^2LID . The authors start from a metric called *border degree* that is used to capture the sharp changes in density near the boundary of rare categories in the feature space:

$$\text{BorderDegree}(\mathbf{x}) = \text{Hub}(\mathbf{x}) - \text{Authority}(\mathbf{x}) \quad (14)$$

where $\text{Hub}(\mathbf{x})$ and $\text{Authority}(\mathbf{x})$ indicate the hub value and authority value [70] of example $\mathbf{x}$, correspondingly. Intuitively, the larger the border degree is, the higher the probability that $\mathbf{x}$ belongs to the rare category. To model task relatedness and view consistency, they then formulate the problem by requiring (1) task-specific learners to behave similarly on the features, (2) view-based learners to behave similarly on the examples, and (3) the border-degree to be negatively correlated with the prediction confidence score. Finally, to solve the optimization problem, they develop a block-coordinate-descent-based method to iteratively update the boundary characteristics of rare categories and multiple classification functions for different tasks. [62] propose a multimodel learning algorithm that is based on a mixture of sparse Gaussian graphical models. [161] aims to tackle the Problem 4 in the presence of data scarcity and adversarial attacks. The authors propose a federated learning framework that performs multiple tasks simultaneously.

The second category is also referred to as transferable rare category exploitation. Instead of learning an optimal predictor for all tasks, transferable rare category exploitation heavily emphasizes the target tasks. Some typical works includes [8], [12], [72], [129], and [136]. In Appel et al. [12], the authors systematically evaluate the performance of anomaly detection frameworks using two types of transfer representation learning paradigms: (1) transfer learning from pre-trained networks, and (2) transfer learning from auxiliary tasks. Kumagai et al. [72] develop an autoencoder framework that learns the domain-specific latent vectors and thus improves the performance in detecting rare examples. Unlike most transfer learning models that aim to learn the transferable representation for the target domain, the work of Vercruyssen et al. [136] proposes to select a subset of labeled and relevant examples from the source domain to augment the target tasks. The aforementioned methods all assume the target domains are related to the source domains. However, there is a more challenging scenario: the so-called *out-of-distribution rare category exploitation*, wherein the test data are unseen or from shifted distributions. To handle the out-of-distribution scenarios, Morteza and Li [98] propose an analytical framework with provable guarantees to characterize and understand out-of-distribution detection in open-world applications. Sun and Wang [128] propose an *adaptive in-out-aware learning framework* to jointly model the mixed in-distribution and out-of-distribution data samples. Acharya et al. [2] propose a graph context reasoning network for out-of-distribution object detection by constructing a representation graph and a context graph from the input image.

4 UNDERSTANDING RARE CATEGORIES

Although they have been tremendously successful in exploring and exploiting rare category examples, many advanced methods, especially deep learning models, often remain black-box in nature. This poses a challenge to their application in industries that have to follow highly regulated processes. In these industries, prediction models must be interpretable and the output results are required to meet compliance standards. Therefore, a natural research question is how to make existing models transparent to the end-user by identifying the right context (e.g., key factors, representative entities, critical timestamps). In this section, we systematically discuss how to

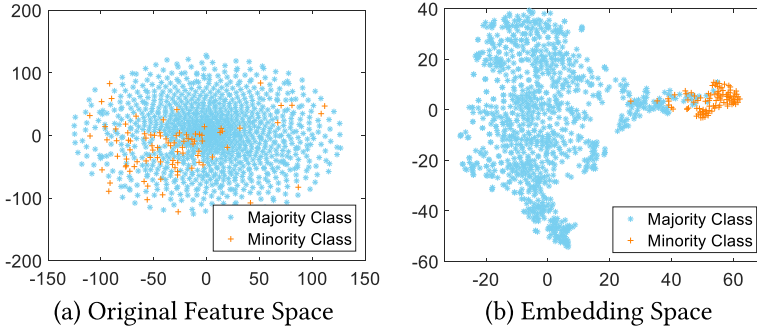


Fig. 9. Rare-category-oriented network representation: the majority and minority classes are not separable in the original feature space, but become well separated in the embedding space [166].

understand rare category patterns from the following two directions: (1) rare category representation and (2) rare category interpretation. The first direction aims to diagnose from the data perspective (i.e., *How is the data distributed? Which piece of information is most valuable for a given task?*), while the second direction aims to diagnose from the model perspective (i.e., *Why does the model make a certain prediction on a particular piece of information?*).

4.1 Rare Category Representation

In this task, we aim to learn a rare-category-oriented embedding representation such that the rare examples (e.g., security threats) are well separated from the majority classes (e.g., normal activities). Figure 9 presents a typical example of rare category representation learning wherein the majority class and the minority class are overlapped together in the original features space but become separated in the learned embedding space. Formally speaking, the problem can be defined as follows:

PROBLEM 5. *Rare Category Representation.*

Given: (i) a dataset $\mathcal{D}$ that consists of n samples; (ii) a small set of labeled examples $\mathcal{D}_l = \{(\mathbf{x}_i)\}_{i=1}^{N_l}$; (iii) the user-defined embedding dimension d .

Find: (i) a rare category representation model $g(\cdot)$; (ii) a d -dimensional embedding representation $\mathbf{E} \in \mathbb{R}^{n \times d}$ that well preserves the underlying distribution of rare categories.

In the past decade, extensive work has been done to develop representation learning methods to learn the underlying distribution of the rare category patterns. The key idea behind these methods is to track the “footprints” of rare categories via external information (e.g., supervision from oracles) or prior knowledge (e.g., features indicating local density changes [52, 147]) and then leverage this information to regularize the representation learning models to extract low-dimensional and salient embedding for rare category analysis. For instance, Akoglu and Falioutsos [5] propose a set of egonet features (e.g., weighted and unweighted in- and out-degree, number of neighbors, number of triangles) from each snapshot of the graph sequence to model the “behavior” of each node. These features are effective in spotting rare category examples. Manzoor et al. [90] aim to spot anomalous patterns in a stream of heterogeneous graphs containing different types of nodes and edges by proposing a clustering-based anomaly detection approach that can simultaneously address the heterogeneity and streaming nature of the input data. In particular, Manzoor et al. introduce a novel embedding mechanism that can encode the heterogeneous streaming graph into a vector representation that is then used to perform clustering and identify anomalous patterns. Huang et al. [60] study the problem of image classification when data in the vision domain exhibit

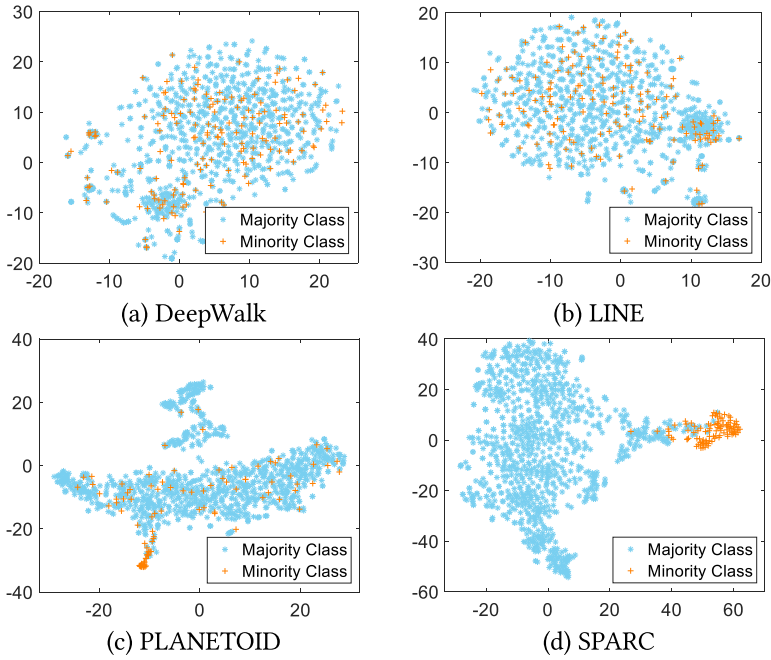


Fig. 10. Preliminary results comparing network embedding using various algorithms [166].

a highly skewed class distribution by learning a deep representation such that rare categories are easily separable from the majority class by contemporary classification methods. In particular, this article first proposes to learn the data representation by maintaining both inter-cluster and inter-class margins that reduce the class imbalance inherent in the local data neighborhood. The proposed framework is built based on a **Convolutional neural network (CNN)** framework through a quintuplet sampling scheme and the associated triple-header hinge loss.

More recently, due to the notable success of network-embedding approaches, various network-embedding-based approaches have been proposed for rare category analysis. For example, *SPARC* [166] is one of the first rare-category-oriented network embedding frameworks. *SPARC* aims to learn a salient representation to characterize rare category examples. Inspired by the family of curriculum learning that simulates the cognitive mechanisms of human beings, *SPARC* gradually selects the key network contextual information and learns the rare-category-oriented network representation by shifting from the “easy” concept to the “difficult” concept. The results show that (1) *SPARC* enables users to visualize the network layout in a salient embedding space, where the majority classes and minority classes are well separated; and (2) *SPARC* is able to identify valuable contextual information that provides interpretation and guidance in the task of rare category characterization. Here we compare *SPARC* with three state-of-the-art network embedding algorithms, including two unsupervised methods, i.e., DeepWalk [111] and LINE [130], and one semi-supervised method, i.e., PLANETOID [148]. We first map the given network to a 129-dimensional space with different embedding methods. Then, we employ the nonlinear dimensionality reduction method, i.e., t-SNE [88], to encode examples in a two-dimensional space for better visualization, as shown in Figure 10. We observe that the rare examples are better clustered using the proposed method than any of the baseline methods.

Following *SPARC*, Wu et al. [142] propose a biased random walk model named *VDRW* for learning imbalanced network representation. *VDRW* is designed to dynamically adjust the transition

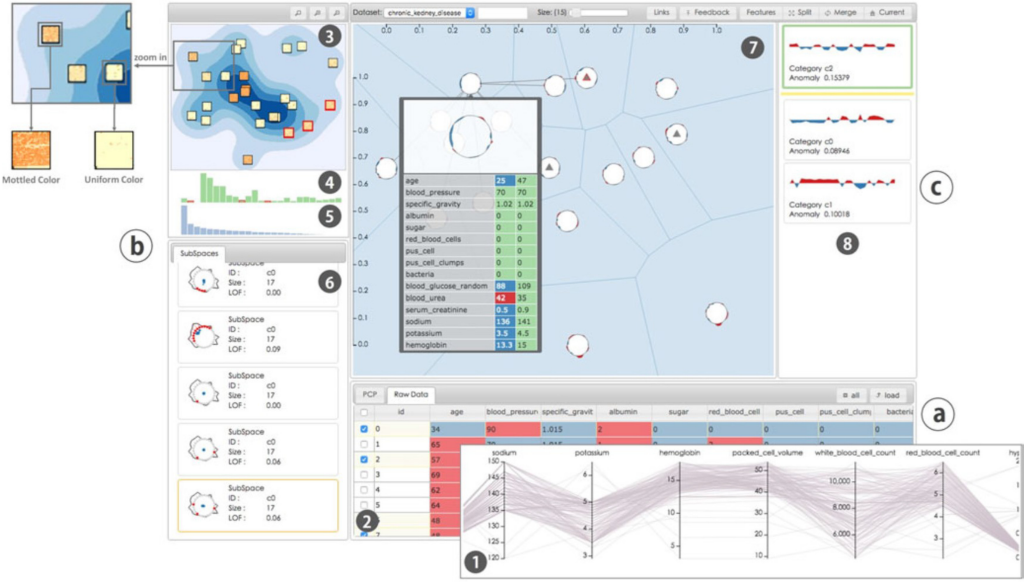


Fig. 11. User interface of *RCLens*, which is composed of three key modules: the data explorer module (shown in subfigure a), the feature explorer module (shown in subfigure b), and the category explorer module (shown in subfigure c).

probability matrix each time a node is visited by the random particle. Via *VDRW*, Wu et al. propose a pair of sampling strategies, i.e., context sampling and balanced-bath sampling, to learn network representation using skip-gram model [94]. *NetWalk* [155] is developed for rare event detection by learning the network representations, which can be updated dynamically as the network evolves. *NetWalk* first extracts the dynamic network context and encodes the vertices of the dynamic network to a low-dimensional representation. Then, a clustering-based scheme is employed to incrementally and dynamically track the malicious patterns in the dynamic networks. Deng et al. [34] propose a GCN-based framework for predicting future events by capturing contextual information from the raw data. The proposed framework first extracts graph representations of the events documents and, then learns to predict the occurrence of future events and identify events of interest (e.g., anomaly patterns).

4.2 Rare Category Interpretation

Interpretability in machine learning models plays a crucial role in many high-impact domains and rare category analysis. In this task, we aim to characterize the rare categories with a predictive model and interpret the output from this model by providing relevant clues, such as connectivity patterns, data sources, features, and timestamps from time-series data. We define the problem setting as follows:

PROBLEM 6. *Rare Category Interpretation.*

Given: (i) a dataset $\mathcal{D}$ that consists of n samples, (ii) a rare category analysis model $f(\cdot)$.

Find: interpretation over the output from $f(\cdot)$ on $\mathcal{D}$.

Despite the rich literature in the context of **explainable artificial intelligence (XAI)** [15, 31, 44, 61, 95, 96, 131], interpreting rare category analysis models remains challenging for the following reasons: (1) rare category analysis models are naturally “biased” (i.e., focusing on minority

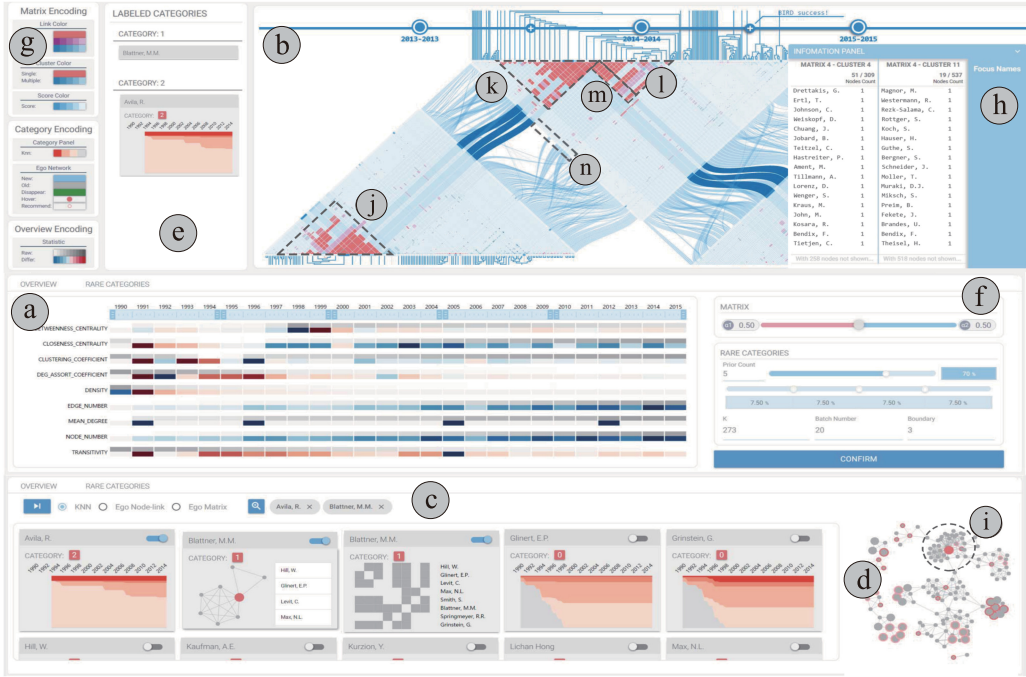


Fig. 12. User interface of RCanalyzer, which is composed of (a) the timeline view; (b) the matrices view; (c) the instance view; (d) the sub-network view; (e) the label result view; (f) the parameter panel; (g) the encoding panel; and (h) the information panel.

patterns instead of the overall data distribution), (2) rare category analysis model $f(\cdot)$ is often trained with scarcity labels, and (3) the learning process of rare category analysis model $f(\cdot)$ may frequently involve oracles and operate in a human-in-the-loop fashion (see Figure 5 for rare category exploration). Simply replacing $f(\cdot)$ with a conventional interpretable machine learning models may deteriorate the prediction performance by providing misleading characterizations or injecting additional bias [74].

There have been a few attempts in the visualization domain to address Problem 6 given the aforementioned challenges. For example, *RCLens* [81] is an interactive visual analytic system that is designed to explore and identify rare category examples with the guidance of end-users. Figure 11 presents the user interface of *RCLens*, which actively visualizes and interprets the three learning phases of rare category analysis models to end-users. In the data exploration phase (shown in subfigure a), the system conducts a personalized visualization of the data via a data querying and filtering mechanism; in the feature selection phase (shown in subfigure a), the system provides relevant clues and statistics (e.g., correlations of features) for end-users to investigate the feature dimensions; and in the rare category analysis model (shown in subfigure c), the system visualizes the identified rare categories to end-users and then refines the predictions by leveraging the users' feedback.

Pan et al. [105] proposes a visual analytic system named RCanalyzer for studying rare category patterns in dynamic systems. The user interface of *RCanalyzer* is shown in Figure 12, which includes (a) the timeline view showing an overview of the given dynamic networks; (b) the matrices view showing the neighborhood contextual information of each node; (c) the example view showing the feature distribution of rare patterns; and (d) the label result view showing the history

Table 2. Publicly Accessible Real-World Datasets for Rare Category Analysis

Data Type	Dataset	Description of Rare Categories
Tabular Data	20NewsGroup [68]	Rare topics of newspapers
Tabular Data	Donors [106]	Outstanding projects proposed by K-12 teachers
Tabular Data	Census [35]	High-income people in U.S. census bureau database
Tabular Data	Fraud [114]	Fraudulent credit card transaction
Tabular Data	UNSW-NB 15 [99]	Network intrusion
Tabular Data	URL [86]	Malicious URLs
Tabular Data	Campaign data [35]	Rarely successful campaigning records
Tabular Data	Thyroid [35]	Rare disease
Time-series Data	Vandal [73]	Wikipedia vandal warning
Time-series Data	Spam [139]	Web spams
Time-series Data	HDFS [145]	System errors
Graph Data	Tox21 [87]	Toxic environmental chemicals
Graph Data	ENZYMES [19]	Abnormal proteins
Graph Data	NCI1 [137]	Abnormal chemical compounds

prediction results as well as the model diagnosis. To ensure the interpretability of rare category analysis models, the system is able to actively visualize each identified rare example in context (e.g., neighborhood structures in the adjacent timestamps) to help oracles understand and examine the prediction results.

5 DATASETS AND REPRESENTATIVE METHODS

This section summarizes the popular evaluation datasets and representative algorithms across different data types and downstream tasks.

5.1 Datasets

Although rare categories are prevalent across various domains (e.g., finance), collecting and annotating rare examples is extremely time-consuming and labor-intensive (e.g., annotating money laundering activities). Early studies (e.g., [48, 50, 52, 53, 109]) evaluate their models on synthetic or semi-synthetic datasets that are either generated from a pre-defined data distribution or manipulated from public benchmark datasets. However, evaluations performed on synthetic or semi-synthetic datasets may not reflect the performance of rare category analysis models in real-world applications. To promote the further development of rare category analysis, Table 2 summarizes some public benchmark datasets, whose data range from tabular data to graph-structured data, from static data to dynamic data.

5.2 Representative Methods

Comparison evaluation with representative algorithms serves a pivotal role in the development of machine learning algorithms. In Table 3, we enumerate and summarize 25 representative rare category analysis algorithms w.r.t. data type (i.e., tabular data, time-series data, graph data) and downstream task (i.e., exploration, exploitation, representation, and visualization). Since the methods in Table 3 have been developed for diverse tasks and datasets, we are unable to provide a meta-analysis of their performance in a unified setting. Instead, to provide insight and an in-depth understanding of the methods, we summarize our major observations as follows: (i) most methods are designed in an unsupervised or semi-supervised setting, (ii) sampling-based methods [135, 163] and anomaly-detection-based methods [132, 158] are still the mainstream

Table 3. Representative Algorithms in Data Perspective

Data Type	Algorithm	Task
Tabular Data	<i>NNDB</i> [52]	Rare Category Exploration
Tabular Data	<i>ACT</i> [141]	Rare Category Exploration
Tabular Data	<i>SEDER</i> [50]	Rare Category Exploration
Tabular Data	<i>HMS</i> [135]	Rare Category Exploration
Tabular Data	<i>MUVIR</i> [163]	Rare Category Exploitation
Tabular Data	<i>RACH</i> [54]	Rare Category Exploitation
Tabular Data	<i>MLAN</i> [104]	Rare Category Exploitation
Tabular Data	<i>RCLens</i> [81]	Rare Category Visualization
Time-series Data	<i>SUITS</i> [80]	Rare Category Exploration
Time-series Data	<i>CDLC</i> [18]	Rare Category Exploration
Time-series Data	<i>MatrixProfile</i> [150]	Rare Category Exploration
Time-series Data	<i>BIRAD</i> [164]	Rare Category Exploration & Exploitation
Time-series Data	<i>RRCF</i> [43]	Rare Category Exploration
Time-series Data	<i>Sand</i> [46]	Rare Category Exploitation
Graph Data	<i>Oddball</i> [6]	Rare Category Exploration
Graph Data	<i>NrMF</i> [132]	Rare Category Exploration
Graph Data	<i>HiDDen</i> [158]	Rare Category Exploration
Graph Data	<i>FocusCO</i> [110]	Rare Category Exploration
Graph Data	<i>BIRD</i> [168]	Rare Category Exploration
Graph Data	<i>StreamSpot</i> [90]	Rare Category Exploration
Graph Data	<i>AnomRank</i> [153]	Rare Category Exploration
Graph Data	<i>SPARC</i> [166]	Rare Category Exploitation & Representation
Graph Data	<i>HOSPLOC</i> [170]	Rare Category Exploitation
Graph Data	<i>L-MEGA</i> [38]	Rare Category Exploitation
Graph Data	<i>RCAnalyzer</i> [105]	Rare Category Visualization

solutions in rare category exploration, and (iii) rare category visualization and interpretation are under explored.

6 CONCLUSION AND FUTURE DIRECTIONS

Departing from the previous surveys, this article presented a comprehensive pipeline of recent advances in rare category analysis. We started from the de-novo step without any label information and surveyed rare category exploration techniques in the setting of homogeneous data and heterogeneous data. Then, we reviewed rare category exploitation methods that aim to characterize a compact representation of the minority classes in order to discover all the rare examples with high accuracy. Finally, we discussed the problem of rare category explanation, and in particular, how to learn a salient representation of rare categories and interpret the prediction results. Despite the significant developments in rare category analysis, there remain some exciting research opportunities

- *Rare Category Generation.* To meet the dramatically increasing demand for machine learning systems as service providers (e.g., social networking, online advertising, data security), a massive amount of data has been generated and collected from a variety of domains. However, the collected rare data are often noisy, sparse, few-shot annotated, and evolving over time. Directly training machine learning models from the raw data would introduce inevitable

model bias and largely degrade the model performance in downstream applications (e.g., rare category characterization, rare category explanation). Moreover, the ever-increasing size of data, together with the difficulty of releasing and sharing them, has made data generation a fundamental problem that is key for many high-impact applications, including fraud detection, recommendation, and data security, among others. Hence, it is critical to develop deep generative models that enable scalable modeling of real data to extract key contextual information, distill knowledge, and generate plausible patterns for data augmentation in rare category analysis.

- *Long-Tail Category Analysis.* In the past decade, deep learning has achieved remarkable success in various learning tasks (e.g., image classification, speech recognition, link prediction) through the training of “big models” on “big data.” However, unlike these well-studied tasks (e.g., image classification over domestic cats and wild cats) with rich training data, the vast majority of real-world entities and patterns (e.g., identification of honest employees and malicious insiders in a large institution) are under explored and lack the observational and annotated data that often correspond to the “long-tail” categories. Unlike existing works on rare category analysis that focus on one or a few rare categories, here we face a massive number of underrepresented categories from a “long-tail” distribution. Moreover, the current machine learning systems are mostly tailored to specific learning scenarios, prevent them from delivering on their promises in detecting the targets of interest in the presence of distribution changes (e.g., dynamic systems). Two fundamental research questions thus emerge: (Q1) *How can we comprehend such massive “long-tail” categories given the inherent paucity of observational and annotated data?* (Q2) *How can we capture the targets of interest (e.g., rare category examples) given a novel data distribution?*
- *Robust Rare Category Analysis.* Robust rare category analysis is another fundamental yet quite open research problem that is attracting a surge of attention from many high-impact domains (e.g., spam detection, financial fraud, system diagnosis). For example, in financial fraud detection: *How can we measure entity sensitivity, algorithmic robustness, task hardness, and model generalization, given a prediction model? How can we achieve operational robustness and adversarial robustness in the presence of the external disturbances (e.g., noise, missing values, outliers, adversarial attacks)?* Despite the extensive work on adversarial machine learning, the vast majority of the previous studies assume a balanced data distribution while neglecting realistic cases in which the data are highly skewed, and the targets of interest are underrepresented. Compared to conventional machine learning tools, rare category analysis models are more sensitive and vulnerable in the presence of adversarial attacks due to the rarity, non-separability, and label scarcity of rare category examples.
- *Human–AI Collaborative Rare Category Analysis.* Many industries follow highly regulated processes that require rare category analysis models to be interpretable and human experts to be involved in the prediction loop. The human-in-the-loop fashion [24, 52] enables rare category analysis models to leverage human intelligence to alleviate the label scarcity issue and improve the prediction performance in high-stake applications, such as failure prediction in safety-critical systems and rare disease diagnosis in health care. However, as human intelligence is naturally a black box to AI models, a fundamental research question remains nascent: *How can we gain a deeper understanding of the underlying mechanism of human intelligence, thus enabling unimpeded knowledge transfer between human intelligence and AI?* A promising but challenging research direction lies in how to build an integrated and interactive system for accurate and trustworthy human–AI collaborative rare category analysis.

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